SCENTINEL

OBP candidate selection · analysis report

iGEM URV 2026

Generated: 18/09/2026 02:06:59

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Target VOC       :  methyl salicylate
CAS              :  119-36-8
Top N candidates :  10
Interferents     :  caryophyllene, β-Cyclocitral, diisobutyl phthalate, 10-undecanal, 1-hexanol, 1-octen-3-ol, acet...
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tau (selectivity):  5
w1 affinity      :  0.450
w2 selectivity   :  0.250
w4 stability     :  0.150
w5 promiscuity   :  0.150
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SMAA configuration

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Method 2 (Perturbation):  c = 50%,  100,000 iterations
Method 3 (pure SMAA)   :  100,000 iterations
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SCENTINEL · Detailed results | VOC: methyl salicylate

#	OBP	Conserv.	Conf.	c2	s1	s2	s4	s5	Ki (uM)	Interf	Danger	Type
1	DcitOBP3	0.4130	0.78	0.16	0.472	0.122	1.000	0.902	7.01	7/43	! 6	Classic OBP
2	GmolOBP7	0.3978	0.82	0.28	0.381	0.088	1.000	0.964	9.81	12/43	! 2	Classic OBP
3	AcerOBP11	0.3925	0.77	0.12	0.479	0.082	1.000	0.814	6.83	5/43	! 4	Classic OBP
4	GmolPBP2	0.3855	0.77	0.12	0.452	0.200	0.850	0.793	7.55	5/43	! 2	PBP
5	AcerOBP2	0.3699	0.75	0.07	0.427	0.066	1.000	0.888	8.29	3/43	! 2	Classic OBP
6	AlinCSP1	0.3668	0.82	0.30	0.414	0.200	0.400	0.994	8.67	13/43	! 1	CSP
7	HoblOBP2	0.3645	0.80	0.21	0.248	0.200	1.000	0.970	16.04	9/43	-	Classic OBP
8	RpadOBP3	0.3642	0.74	0.07	0.350	0.162	1.000	0.951	11.00	3/43	! 2	Classic OBP
9	SmosOBP12	0.3611	0.77	0.12	0.386	0.110	1.000	0.780	9.62	5/43	! 4	Classic OBP
10	PxylGOBP2	0.3554	0.79	0.19	0.437	0.025	0.850	0.795	7.97	8/43	! 7	GOBP2

Conserv. = Score x Confidence · c2 = selectivity confidence · Interf = measured/total · Danger = vineyard interferents binding as strong as the target
Row tint: green = confidence >= 0.80 lilac = 0.50-0.80 pink = < 0.50

Dangerous vineyard interferents per candidate

(Ki_interf <= Ki_target x 1.33; v = binds stronger than the target)

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 1. DcitOBP3         Ki=  7.01 [6]  C9:Ald(4.3v), acetic acid(5.3v), γ-Terpinene(6.8v), r-limonene(8.2), caryophyllen...
 2. GmolOBP7         Ki=  9.81 [2]  12:Ald(4.3v), 10:Ald(12.7)
 3. AcerOBP11        Ki=  6.83 [4]  1-hexanol(2.8v), phenylethanal(3.8v), Ethyl hexadecanoate(4.4v), geraniol(8.6)
 4. GmolPBP2         Ki=  7.55 [2]  C9:Ald(7.6), hexanal(8.0)
 5. AcerOBP2         Ki=  8.29 [2]  phenylethanal(2.7v), geraniol(5.5v)
 6. AlinCSP1         Ki=  8.67 [1]  caryophyllene(9.2)
 8. RpadOBP3         Ki= 11.00 [2]  beta-farnesene(8.9v), r-limonene(10.5v)
 9. SmosOBP12        Ki=  9.62 [4]  1-octen-3-ol(5.3v), 2-ethyl-1-hexanol(6.4v), 12:Ald(8.1v), Ocimene(8.2v)
10. PxylGOBP2        Ki=  7.97 [7]  Farnesene(1.0v), caryophyllene(2.4v), phenylethanal(4.2v), geraniol(5.5v), 1-hexa...
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SCENTINEL · s5 superbinders | VOC: methyl salicylate

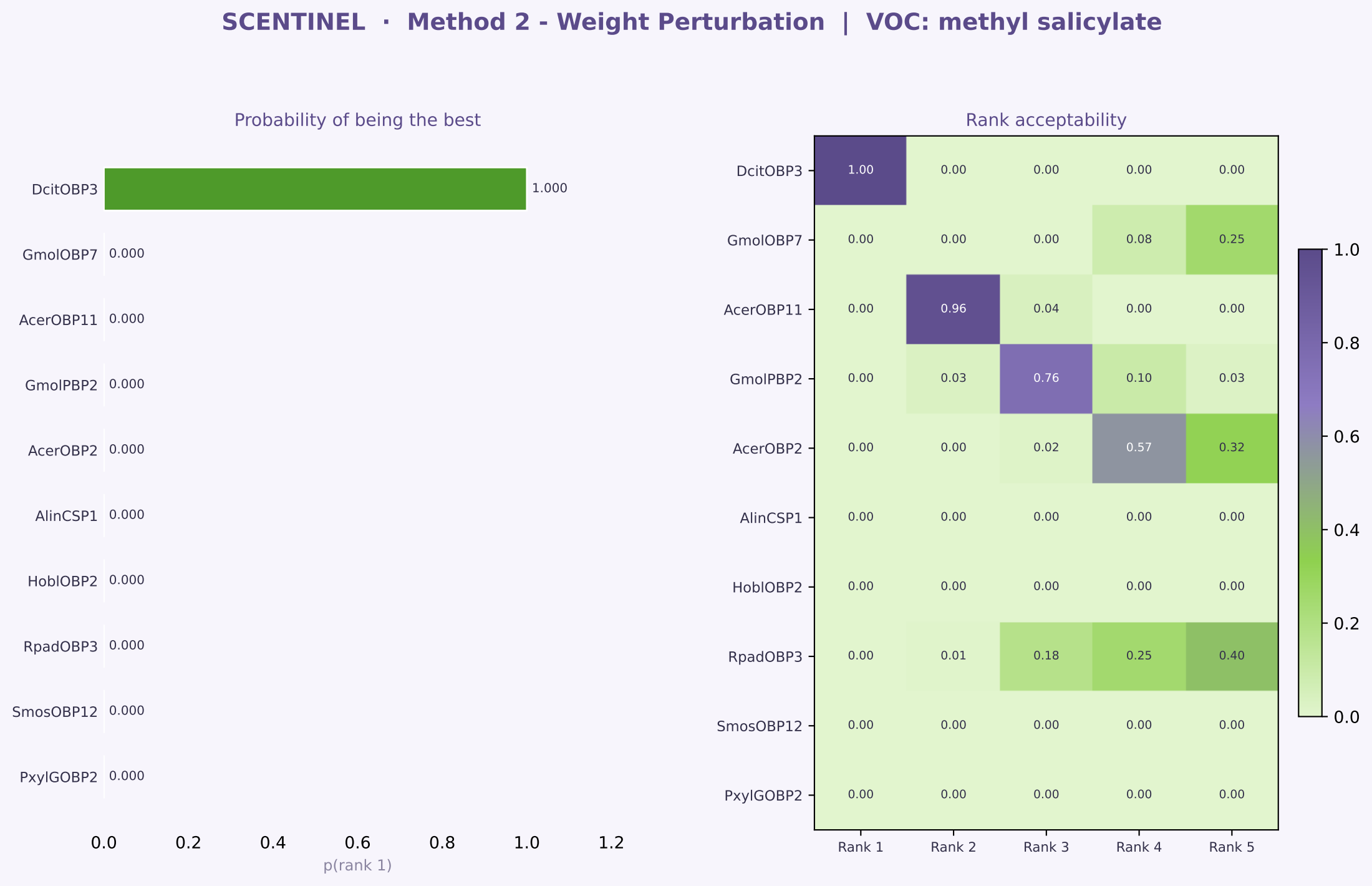
Superbinders - VOCs (any, not only vineyard) bound strongly

* = Ki <= 2.0 uM (strong) v = binds stronger than the target [n] = total superbinders

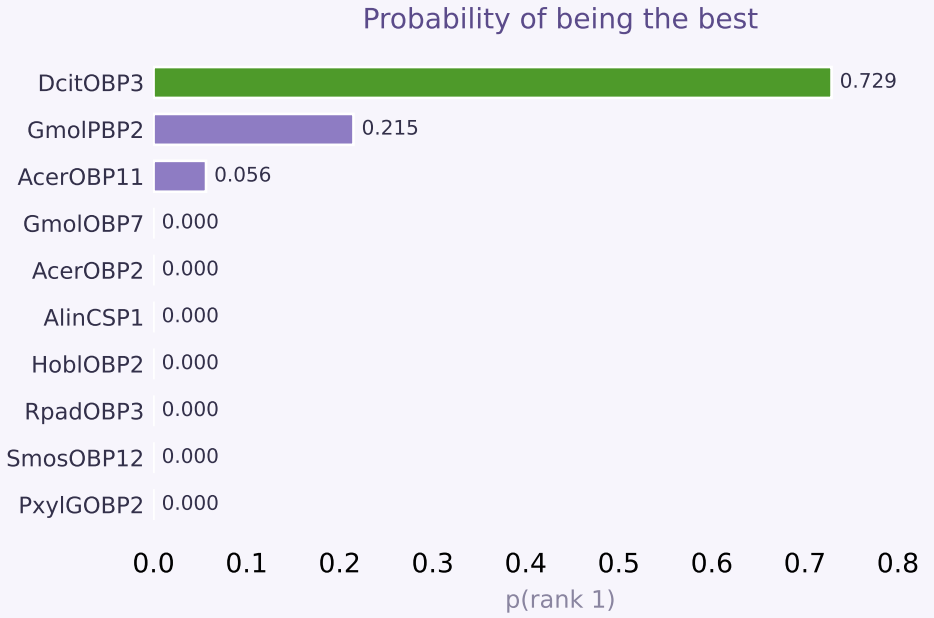
1. DcitOBP3 [11] DL linalool(3.5v), dimethyl trisulfide(4.2v), C9:Ald(4.3v), p-cymene(4.7v), (1R)-(+)-alpha-pinene(5.0v), (1S)-(-)-alpha-pinene(5.1v), dimethyl disulfide(5.1v), acetic acid(5.3v), trans-(3E)-beta-ocimene(5.9v), lauric acid(6.4v) +1
2. GmolOBP7 [4] Pear ester(2.5v), 12:Ald(4.3v), 12:OH(7.5v), alpha-ocimene(7.8v)
3. AcerOBP11 [16] HOB(1.4*v), Methyleugenol(2.7v), 1-hexanol(2.8v), isoamyl acetate(3.3v), phenylethanal(3.8v), ionone (beta)(3.8v), Ethyl hexadecanoate(3.8v), eucalyptol(4.3v), 2-heptanone(4.3v), l-(-)-Menthol(4.3v) +6
4. GmolPBP2 [8] Z8-12:Ac(1.1*v), E8-12:AC(1.1*v), Z8-12:OH(2.0*v), alpha-pinene(2.0v), 12:OH(2.8v), heptanal(3.8v), Cis-3-Hexenyl acetate(5.8v), ionone (beta)(6.7v)
5. AcerOBP2 [4] phenylethanal(2.7v), ionone (beta)(4.1v), geraniol(5.5v), HOB(7.6v)
6. AlinCSP1 [3] myrcene(5.4v), beta-pinene(7.1v), Cis-3-Hexenyl acetate(7.4v)
7. HoblOBP2 [4] ionone (beta)(5.4v), Hexyl benzoate(8.0v), ionone (alpha)(10.4v), Butyl benzoate(16.0v)
8. RpadOBP3 [4] beta-farnesene(8.9v), Z-3-hexen-1-ol(9.2v), (1S)-(-)-alpha-pinene(9.7v), r-limonene(10.5v)
9. SmosOBP12 [16] 3-hexanol(3.3v), heptyl acetate(3.3v), 3-Carene(3.8v), D-panthenol(4.1v), Pentyl pentanoate(4.2v), 1-undecanol(4.4v), 1-octen-3-ol(5.3v), propanoic acid, hexyl ester(5.9v), alpha-pinene(6.2v), 3-hexanone(6.3v) +6
10. PxylGOBP2 [17] Farnesene(1.0*v), (Z)-3-Hexenyl isovalerate(2.4v), Z11-16:Ac(2.4v), caryophyllene(2.4v), Z11-16:OH(2.7v), Z11-16:Ald(2.9v), phenylethanal(4.2v), DL linalool(4.5v), benzoic acid(4.9v), geraniol(5.5v) +7

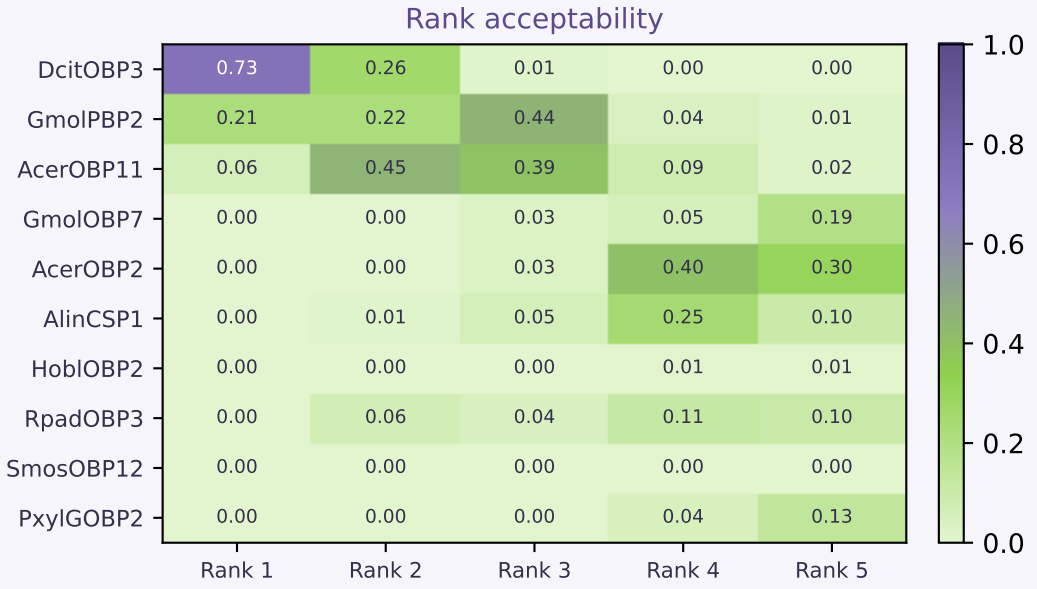
SCENTINEL · Experimental vs Computational | VOC: methyl salicylate

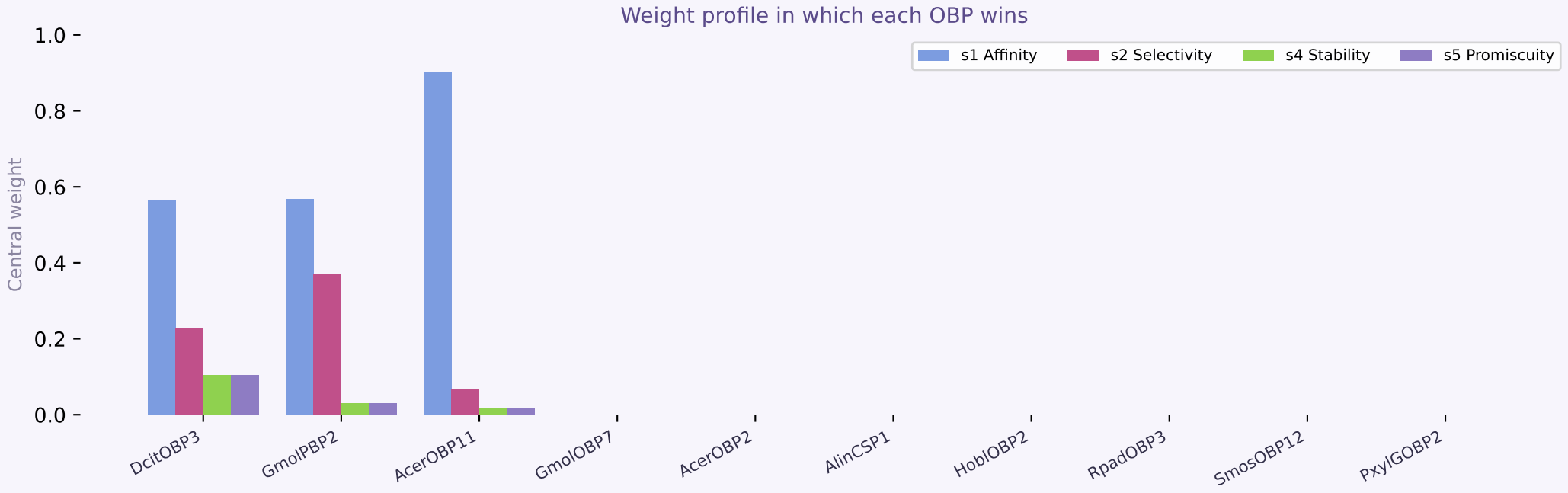



SCENTINEL · Method 2 - Weight Perturbation | VOC: methyl salicylate
Probability of being the best
DcitOBP3
GmolOBP7
AcerOBP11
GmolPBP2
AcerOBP2
AlinCSP1
HoblOBP2
RpadOBP3
SmosOBP12
PxylGOBP2
1.000
0.000
0.0
0.2
0.4
0.6
0.8
1.0
1.2
p(rank 1)
Rank acceptability
1.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.08 0.25
0.00 0.96 0.04 0.00 0.00
0.00 0.03 0.76 0.10 0.03
0.00 0.00 0.02 0.57 0.32
0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00
0.00 0.01 0.18 0.25 0.40
0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00
Rank 1
Rank 2
Rank 3
Rank 4
Rank 5
1.0
0.8
0.6
0.4
0.2
0.0

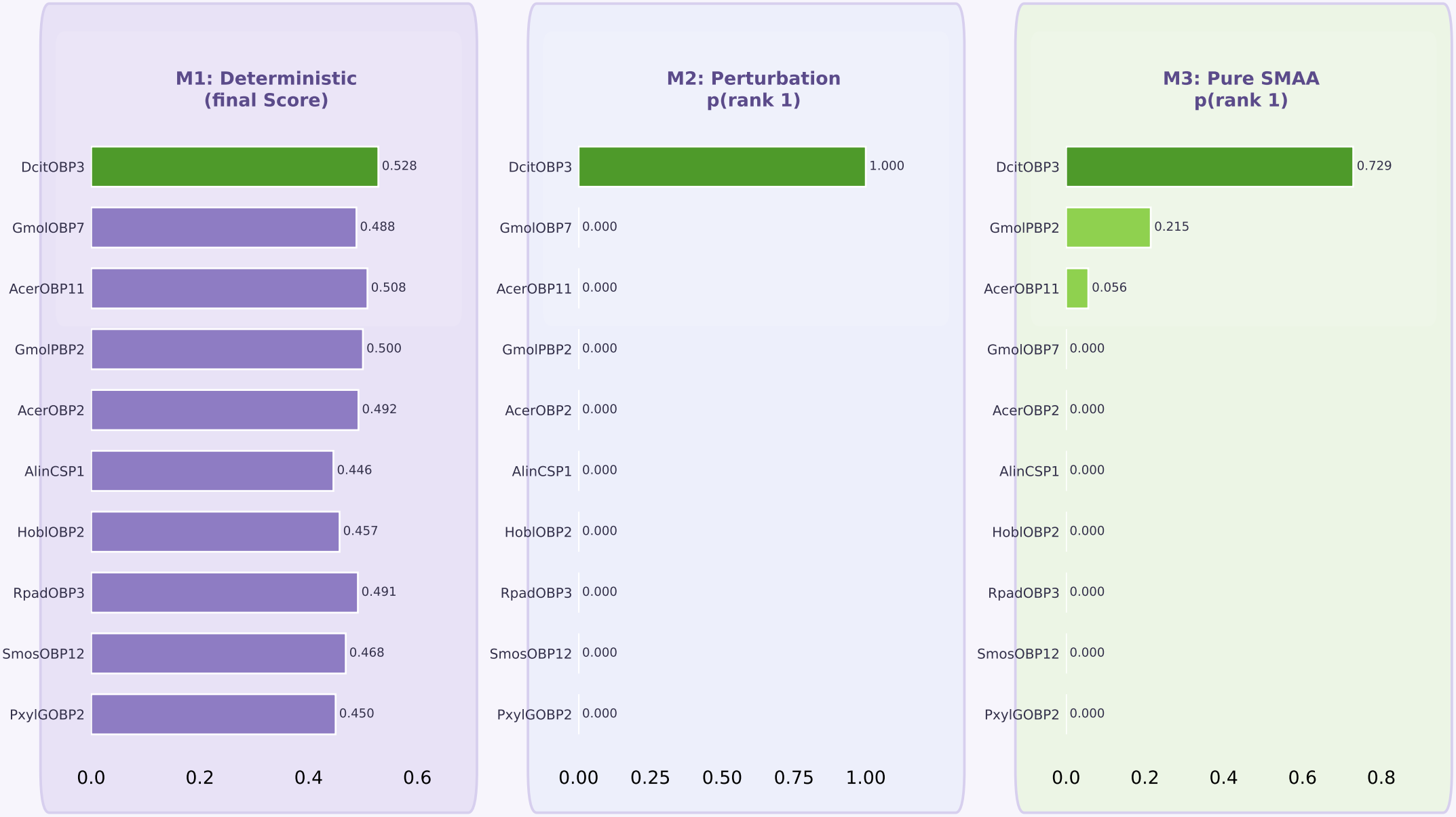
SCENTINEL · Method 3 - Pure SMAA | VOC: methyl salicylate







SCENTINEL · Comparison of the 3 methods | VOC: methyl salicylate



Winner · M1 (deterministic): DcitOBP3 M2 (perturbation): DcitOBP3 M3 (pure SMAA): DcitOBP3